

Drones Scrambled

Developing an *aatmanirbhar* ecosystem of products and services is vital to becoming a global drone manufacturing hub

BY **MANISH PANT**

► **EVEN AS THE CIVIL AVIATION MINISTRY** looks to push India's budding drone sector to become a \$30-billion industry by 2030, disruptions caused by pandemic-induced lockdowns and geopolitical risks have led domestic players to seek self-reliance.

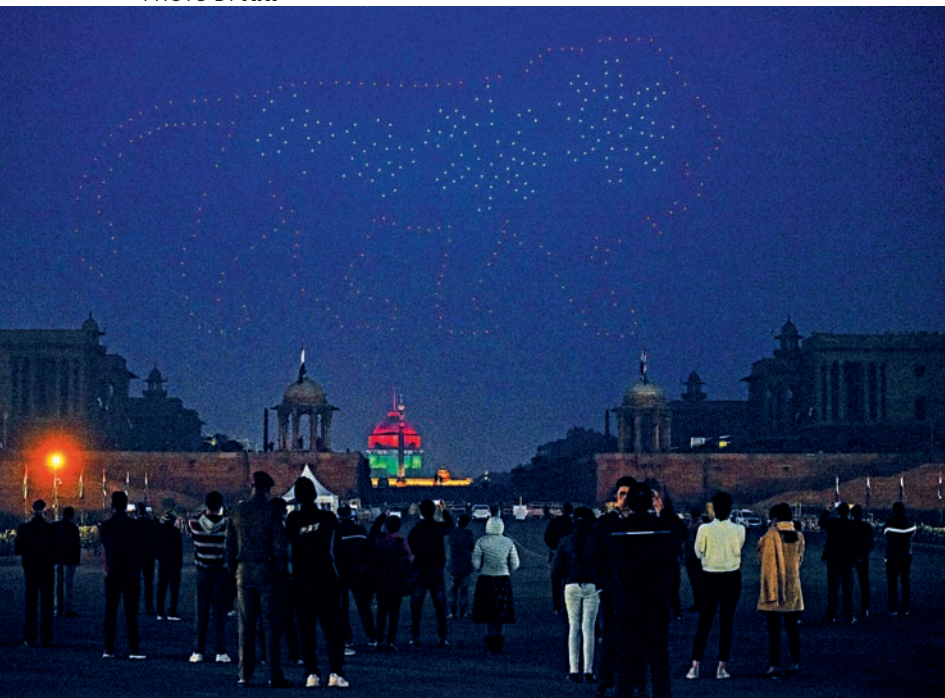
One area hit hardest by these headwinds is the supply of lithium-polymer and lithium-ion batteries, resulting in manufacturers migrating to graphene-based cells, whose weight weighs on the drones' efficiency.

"We explored solar but it results in a trade-off with the payload capacity. There needs to be support on battery technologies, where we can produce newer batteries in India," says Samriddhi Pandey, Founder and CEO of drone-making start-up Defy Aerospace.

Similarly, Indian manufacturers are reliant on Chinese imports for avionics such as motors and flight controllers. "It's difficult to get them from India. The ones we have come from the US and are very expensive, taking away our cost advantage," informs Pandey.

**A DAZZLING
DISPLAY OF DRONE
FORMATION AT
THE BEATING
RETREAT
CEREMONY AT
VIJAY CHOWK, IN
NEW DELHI**

PHOTO BY **ANI**



"The harsh truth is, we are dependent on battery imports," says Amber Dubey, Joint Secretary in the Ministry of Civil Aviation. "Import substitution requires growth-oriented policies, production incentives, and demand creation. Investments, infrastructure and supply chain will automatically follow."

A long-term approach will involve attaining self-reliance with everyone working towards getting rid of import dependence, suggests Vipul Singh, Co-founder and CEO of AUS, an enterprise drone maker. "There is a gap in terms of creating intellectual property (IP). If we don't do that, we will create a manufacturing hub with a lot of blue-collar jobs, but the wealth will flow where the IP resides," he warns.

The ministry, meanwhile, emphasises its collaborations—with start-ups, academia and various government departments—are aimed at boosting the domestic drone industry. As the production-linked incentive scheme funnels financing to battery manufacturers, the ban on drone imports aims at helping domestic start-ups. "The massive demand for agriculture, survey, surveillance, cargo and military use cases is likely to attract investments in battery design and manufacturing. We hope to achieve *aatmanirbharta* (self-reliance) in drone batteries over the next five years," asserts Dubey.

Some companies have started working with local manufacturers to fill the gap. "Domestic companies have... started work on battery assembly. We are working on R&D to help drive the indigenisation of components," says Nihar Vartak, Co-founder of Asteria Aerospace, which designs military and enterprise drones.

For drones weighing less than 2 kg, it sources batteries from local suppliers that give the same level of performance as imported ones. These drones are now successfully conducting enterprise activities. **BT**

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